

CoRoT-1b

R-2575

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_180201 · created 2026-08-27T00:16:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458150.6807 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.043
Transit depth (Rp/Rs)^2	2.04 +/- 1.22 [%]
Orbital Inclination (inc)	81.5 +/- 2.9
Airmass coefficient 1 (a1)	1.002 +/- 0.043
Airmass coefficient 2 (a2)	-0.0 +/- 0.024
Scatter in the residuals of the lightcurve fit is	4.36 %
Best Comparison Star	#2 - [162, 217]
Optimal Aperture	2.64
Optimal Annulus	12.1
Transit Duration (day)	0.084 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.043 vs 0.1388 (deviation 0.07σ)
Duration fit vs published	0.084 d vs 0.1046 d (deviation 0.58σ)
Mid-time O–C	8.1 min
Depth SNR	2.4
Residual scatter	4.36 %

Light curve

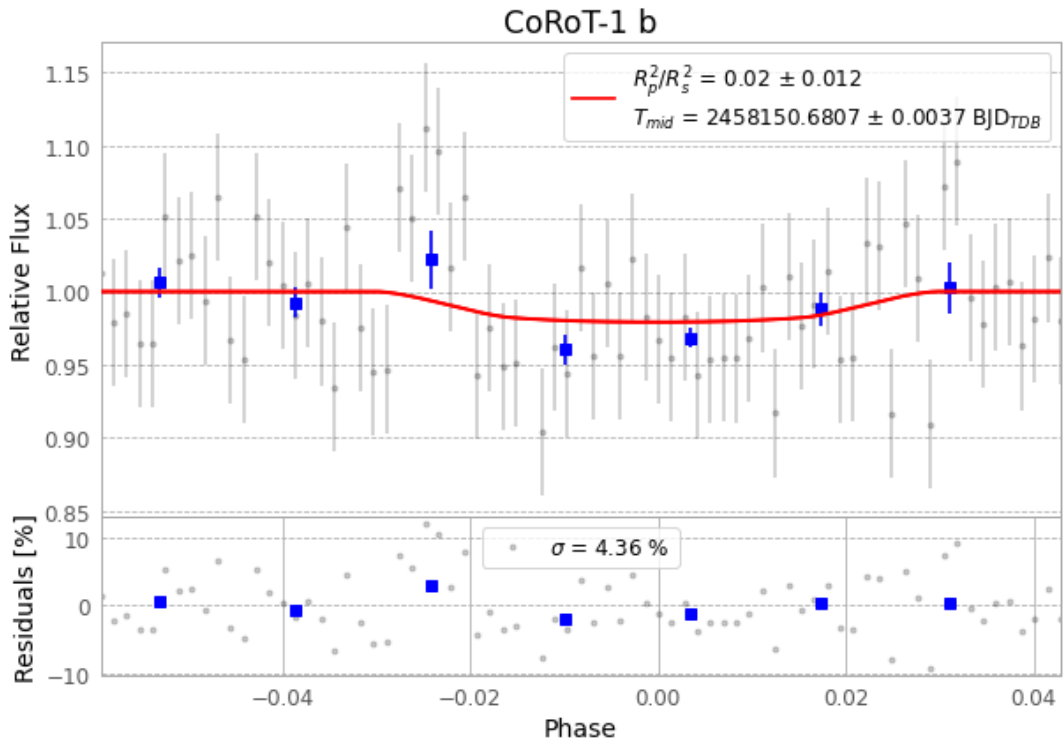


Figure 1 — FinalLightCurve_CoRoT-1 b_2018-01-31.png (SHA-256 18596485c2498b2a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 197], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bf5cf9e3677cd6e4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), CoRoT-1180201020312.FITS → CoRoT-1180201054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-180201.log (b191f3502b3e...), FinalLightCurve_CoRoT-1 b_2018-01-31.png (18596485c249...), FinalLightCurve_CoRoT-1 b_2018-01-31.csv (720e0abff8ef...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)